Sequence 10

VARIATIONS AND EXTREMA OF A FUNCTION

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I Increasing, decreasing and monotonicity

Definitions

On an interval I, f is increasing if for all $a, b \in I$, $a < b \Rightarrow f(a) \leq f(b)$; decreasing if $a < b \Rightarrow f(a) \geq f(b)$. The variations are strict when the inequalities between images are strict. A constant function has the same value everywhere; it is both increasing and decreasing in the broad sense. A function that is monotonic on I is increasing or decreasing there.

Table of variations

Indicate the domain, the intervals of monotonicity, the values at the endpoints when defined, then the arrows. A sign table shows whether $f(x)$ is positive or negative; a table of variations shows how its values change.

Example 10.1

On $[-3;4]$, $f(x) = (x-1)^2 - 2$. Using the square, extending the curve studied in sequence 6, draw its table of variations. Give two numbers $a < b$ such that $f(a) > f(b)$.

Solution:

The function decreases strictly on $[-3;1]$, then increases strictly on $[1;4]$. $f(-3) = 14$, $f(1) = -2$, $f(4) = 7$.

x	-3		1		4
$f(x)$	14	$\searrow$	-2	$\nearrow$	7

For example $a = 0 < b = 1$ and $f(0) = -1 > -2 = f(1)$.

Example 10.2

Can a function be negative and increasing? Give an example on an interval.

Solution:

Yes: $f(x) = x - 5$ on $[0;2]$ is strictly increasing and stays between -5 and -3, so it is strictly negative.

II Maximum and minimum on an interval

Definitions

f has a maximum at $a \in I$ if, for every $x \in I$, $f(x) \leq f(a)$. The value of the maximum is $f(a)$. f has a minimum at $b \in I$ if, for every $x \in I$, $f(x) \geq f(b)$. An extremum must be attained. Distinguish its value from the abscissa where it is attained.

Example 10.3

Return to $f(x) = (x-1)^2 - 2$ on $[-3; 4]$. Give the minimum and maximum, then where they are attained. Repeat on $[0; 4]$.

Solution:

On $[-3; 4]$, minimum -2 at $x = 1$, maximum 14 at $x = -3$. On $[0; 4]$, the minimum remains -2 at 1, but the maximum is 7 at 4. The maximum depends on the domain.

Example 10.4

Does the function $f(x) = x$ have a maximum on $[0; 1[$? A minimum?

Solution:

It has minimum 0, attained at 0. It has no maximum: for every $x < 1$, $(x+1)/2$ still belongs to $[0; 1[$ and has a larger image. The number 1 is an upper bound, but is not attained.

Reading without inventing

A table of variations does not provide all the intermediate images. It allows comparing certain images; if the abscissae lie on different branches, the comparison may be indeterminate.

Example 10.5

Does the mere fact that a function decreases then increases allow comparing two images on either side of the minimum?

Solution:

No. Additional information is needed. For $x \mapsto x^2$, the images of -2 and 1 are 4 and 1, while those of -1 and 2 are 1 and 4.

III Variations of affine functions

Property and rate of change

For $f(x) = mx + p$ and $a \neq b$,

$$\frac{f(b) - f(a)}{b - a} = m.$$

If $m > 0$, f is strictly increasing; if $m < 0$, strictly decreasing; if $m = 0$, constant.

Proof to write.

Solution:

For $a < b$, $f(b) - f(a) = m(b - a)$. Since $b - a > 0$, the difference has the sign of m. It is positive, negative or zero according to the three cases, which gives the stated variations. Dividing by $b - a$ gives the constant rate of change.

Example 10.6

Determine the variations of $f(x) = 3x + 2$, $g(x) = 7 - 6x$, $h(x) = -x$. Give the sign of $2x - 6$ and explain why the sign of this function does not stay constant although it is increasing.

Solution:

f is strictly increasing; g, h strictly decreasing. $2x - 6$ is negative for $x < 3$, zero at 3, positive for $x > 3$. An increasing function can cross 0; its direction of variation does not determine a single sign.

Example 10.7

An affine function u satisfies $u(2) = 5$, $u(6) = 13$. Determine u, then its minimum and maximum on $[-1; 4]$.

Solution:

$m = (13 - 5)/(6 - 2) = 2$, $p = 1$, so $u(x) = 2x + 1$. It is increasing: minimum $u(-1) = -1$, maximum $u(4) = 9$.

IV Variations of the square and the absolute value

Properties

$x \mapsto x^2$ and $x \mapsto |x|$ are strictly decreasing on $]-\infty;0]$ and strictly increasing on $[0;+\infty[$. Their minimum on $\mathbb{R}$ is 0, attained only at 0.

Proof to write.

Solution:

For $a < b$, $b^2 - a^2 = (b-a)(a+b)$. If $0 \leq a < b$, both factors are strictly positive: $a^2 < b^2$. If $a < b \leq 0$, the first factor is positive and the second negative: $a^2 > b^2$. For the absolute value, on $[0;+\infty[$, $|x| = x$ is increasing; on $]-\infty;0]$, $|x| = -x$ is decreasing. Positivity and vanishing at 0 give the minimum.

Example 10.8

Compare without a calculator: $(-3.2)^2$ and $(-3.1)^2$; 1.2^2 and 1.3^2; $|-5|$ and $|-2|$.

Solution:

On the negatives, the square reverses the order: $(-3.2)^2 > (-3.1)^2$. On the positives, $1.2^2 < 1.3^2$. Finally $|-5| > |-2|$, that is $5 > 2$.

Example 10.9

Bound x^2 if $-3 \leq x \leq 2$. Why can we not simply square the two bounds?

Solution:

The minimum is 0 and the maximum 9: $0 \leq x^2 \leq 9$. The square is not monotonic on $[-3;2]$. Squaring only the bounds would wrongly give $9 \leq x^2 \leq 4$.

V Variations of the inverse function

Property

The inverse function is strictly decreasing on $]-\infty;0[$ and on $]0;+\infty[$, separately. It is not decreasing on the whole of $\mathbb{R}^*$.

Proof to write.

Solution:

For $a < b$ of the same sign, $ab > 0$ and

$$\frac{1}{b} - \frac{1}{a} = \frac{a-b}{ab} < 0.$$

So $1/b < 1/a$. On the other hand, $-1 < 1$ and $1/(-1) = -1 < 1 = 1/1$: the decreasing criterion fails on the whole of $\mathbb{R}^*$.

Example 10.10

Compare $1/(-4)$ and $1/(-2)$, then $1/3$ and $1/5$. Bound $1/x$ if $2 \leq x \leq 5$, then if $-5 \leq x \leq -2$.

Solution:

$-1/4 > -1/2$; $1/3 > 1/5$. For $x \in [2;5]$, $1/5 \leq 1/x \leq 1/2$. For $x \in [-5;-2]$, $-1/2 \leq 1/x \leq -1/5$.

Example 10.11

Why is the interval $[-1;1]$ not suitable for directly applying a variation rule of the inverse function?

Solution:

The function is not defined at 0, which belongs to the interval. One must work on the two parts of the domain; moreover, the comparison behaviour of the function is not the same when crossing 0.

VI Relative position of a number and its square

Property on the positives

For $0 \leq x \leq 1$, $x^2 \leq x$; for $x \geq 1$, $x^2 \geq x$. Equality holds at 0 and 1.

Proof to write.

Solution:

$x^2 - x = x(x-1)$. On $[0;1]$, the first factor is positive or zero, the second negative or zero: their product is negative or zero. On $[1;+\infty[$, both are positive or zero. The product is zero if $x = 0$ or $x = 1$.

Square root and cube

The square root function is strictly increasing on $[0;+\infty[$. The cube function is strictly increasing on $\mathbb{R}$.

Example 10.12

Explain why the square root is increasing from the increase of the square on the positives. For the cube, use $b^3 - a^3 = (b-a)(a^2+ab+b^2)$.

Solution:

If $0 \leq a < b$ but $\sqrt{a} \geq \sqrt{b}$, the increase of the square on the positives would give $a \geq b$, a contradiction. For $a < b$, $b - a > 0$ and $a^2 + ab + b^2 = (a + b/2)^2 + 3b^2/4 > 0$ (equality would require $a = b = 0$). So $b^3 - a^3 > 0$.

Example 10.13

Compare $\sqrt{5}$ and $\sqrt{6}$, then $(-2)^3$ and $(-1)^3$.

Solution:

$\sqrt{5} < \sqrt{6}$ and $-8 < -1$, consistent with the two increasing behaviours.

VII Optimising: conjecture and prove

Example 10.14

A rectangle of perimeter 20 m has area $A(x) = x(10 - x)$ with $x \in]0; 10[$.

1. Conjecture the maximum area using a table of values.
2. Show that $A(x) = 25 - (x - 5)^2$. Deduce the exact maximum and the corresponding dimensions.

Solution:

For $x = 1, 2, 3, 4, 5, 6, 7, 8, 9$, the areas are $9, 16, 21, 24, 25, 24, 21, 16, 9$ m^2. Expanding, $25-(x-5)^2 = 10x - x^2 = A(x)$. Since $(x - 5)^2 \geq 0$, $A(x) \leq 25$, with equality only at $x = 5$. The optimal rectangle is a square of side 5 m.

```
def aire(x):
    return x*(10-x)

def maximum_grille(N):
    # N entier >= 2 ; points strictement entre 0 et 10
    meilleur_x = 10/N
    meilleur_y = aire(meilleur_x)
    for k in range(2, N):
        x = 10*k/N
        if aire(x) > meilleur_y:
            meilleur_x, meilleur_y = x, aire(x)
    return meilleur_x, meilleur_y
```

Example 10.15

What does the program return for $N = 10$? Does a numerical sweep always prove the maximum of a function on an interval?

Solution:

For $N = 10$, $(5, 25)$. The program finds the maximum only over the sampled points. It can miss a maximum between two points: the algebraic proof gives the exact result here. An ordinary bisection for finding a zero does not automatically apply to finding an extremum.